Truncated Regression

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Introduction

Truncated regression applies when observations outside a range are absent entirely from the dataset rather than being censored at the threshold. The distinction matters: censored data record a value at the threshold for observations that fall outside, while truncated data simply omit them. Examples: a study limited to hypertensive patients (no normotensive observations), an income survey of households above a threshold (no low-income data), a registry of high-grade tumours (no low-grade entries). Ordinary least squares on truncated data is biased because the conditional mean inside the truncation region is non-linear, even when the latent model is linear.

Prerequisites

A working understanding of linear regression, the difference between censoring and truncation, and the truncated-Normal distribution.

Theory

For latent $Y^* = X^\top \beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, truncation at L from below means we observe Y only when $Y^* > L$. The conditional density on the truncated region is

$$f(y \mid x, Y > L) = \frac{\phi((y - x^\top \beta)/\sigma)/\sigma}{1 - \Phi((L - x^\top \beta)/\sigma)}.$$

The conditional mean inside the truncation region depends non-linearly on x via the truncation point, so OLS coefficients are attenuated. Maximum-likelihood estimation under the correct truncated-Normal density recovers the latent-model coefficients. The corresponding model with covariate-dependent truncation rules is the Heckman selection model (`sampleSelection::selection`).

Assumptions

A known truncation threshold (or a known truncation rule), Normal latent errors, correctly specified latent linear model. Heavy truncation makes inference fragile because few observations remain.

R Implementation

```r
library(truncreg)

set.seed(2026)
x_all <- rnorm(300)
y_all <- 2 + 1.5 * x_all + rnorm(300)
keep  <- y_all > 0                 # truncation: keep only y > 0
x <- x_all[keep]; y <- y_all[keep]

fit_trunc <- truncreg(y ~ x, point = 0, direction = "left")
summary(fit_trunc)
```

```r
# Biased OLS
coef(lm(y ~ x))
```

Output & Results

`truncreg()` returns coefficient estimates that recover the latent-model parameters (close to the data-generating values 2 and 1.5). Naive OLS on the same truncated data attenuates the slope toward zero, illustrating the bias.

Interpretation

A reporting sentence: "Truncated regression accounting for the selection rule ($y > 0$) recovered the slope at 1.48 (95 % CI 1.20 to 1.76), close to the data-generating value of 1.5; naive OLS on the same truncated sample produced an attenuated slope of 0.91 (95 % CI 0.65 to 1.17), demonstrating the consequence of ignoring truncation." Always check whether observations were dropped before reaching the analysis dataset; truncation in the data-collection process is invisible to standard regression diagnostics.

Practical Tips

- Truncation is stronger than censoring: truncated values are unobserved, censored values are observed at the boundary. Use the right model for the data structure.
- For covariate-dependent truncation rules, Heckman two-step or `sampleSelection::selection()` jointly models the selection and outcome equations.
- Truncated regression is sensitive to the Normality assumption; check residuals on the truncated scale and consider semi-parametric alternatives if Normality fails.
- For heavily truncated samples (most of the latent distribution unobserved), inference becomes fragile; report the proportion of latent observations retained.
- Bayesian alternatives with informative priors stabilise truncated fits when the data alone are weak.
- Confirm the truncation point (often a study design parameter) before fitting; misstating L produces systematically biased estimates.

R Packages Used

`truncreg` for the canonical truncated-Normal regression; `sampleSelection` for Heckman two-step and ML-based sample-selection models; `crch` for censored / truncated regression with heteroscedastic errors; `gamlss` for truncated outcomes within a richer distributional-regression framework.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors

- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures $\rightarrow$ mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI